

Lab - Continuous Integration with Jenkins

In this lab you will work with Jenkins which is a Continuous Integration (**CI**) tool.

The project supplied is written in Java, but Jenkins' CI pipeline can work for any buildable software.

Objectives

- To get Jenkins up and running using Docker
- To explore the options Jenkins gives us for connecting to other services
- To explore the Jenkins dashboard

Prerequisites

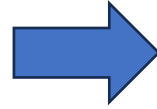
- Docker desktop is installed
- Git CLI
- You have a GitHub account



Objective of this exercise



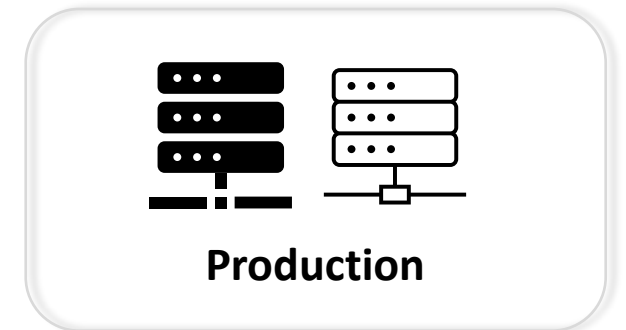
Write code
Make changes
Check the code in



Detects the change
Raises a build event



Build and runs tests
Deploy the code
Report result to GitHub



In this lab, you will

- Deploy a Maven app from GitHub to Jenkins
- Build and test the app using Jenkins
- Make changes to your code and deploy the changes to GitHub
- Set up Jenkins to detect the change in code and rebuild the app

Run Jenkins using Docker

In a terminal window, type

```
docker run -p 8080:8080 -p 50000:50000 -v jenkins_home:/var/jenkins_home jenkins/jenkins:lts-jdk11
```



```
cmd: Select Administrator: Command Prompt - docker run -p 8080:8080 -p 50000:50000 -v jenkins_home:/var/jenkins_home
2023-07-14 18:09:37.314+0000 [id=30] INFO jenkins.InitReactorRunner$1#onAttache
2023-07-14 18:09:37.316+0000 [id=30] INFO jenkins.InitReactorRunner$1#onAttache
2023-07-14 18:09:37.320+0000 [id=29] INFO jenkins.InitReactorRunner$1#onAttache
2023-07-14 18:09:37.425+0000 [id=44] INFO hudson.util.Retrier#start: Attempt 1
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.codehaus.groovy.vmplugin.v7.Java7$1 (file:/var/jenkins_home/groovy-all-2.4.21.jar) to constructor java.lang.invoke.MethodHandles$Lookup(java.lang.Class)
WARNING: Please consider reporting this to the maintainers of org.codehaus.groovy.
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective
WARNING: All illegal access operations will be denied in a future release
2023-07-14 18:09:38.816+0000 [id=28] INFO jenkins.install.SetupWizard#init:

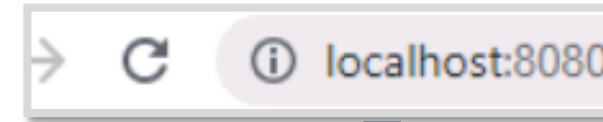
*****
*****
*****

Jenkins initial setup is required. An admin user has been created and a password generated.
Please use the following password to proceed to installation:

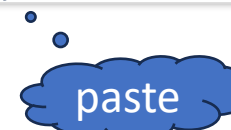
2cec52a3bda045fcbd55204f578d775e

This may also be found at: /var/jenkins_home/secrets/initialAdminPassword

*****
*****
*****
```



Run Jenkins in a browser





Install suggested plugins

Install plugins the Jenkins community finds most useful.



Create First Admin User

Username

Password

Confirm password

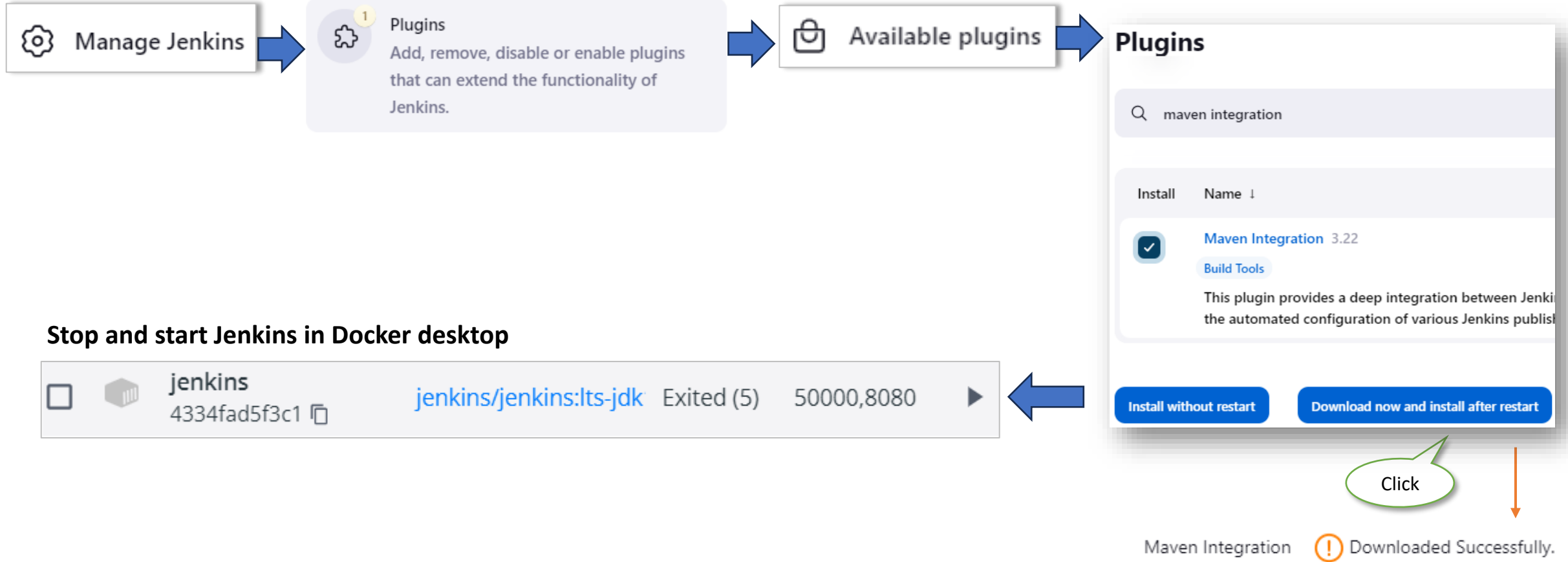
Full name



Instance Configuration

Jenkins URL:

Accept or
change the port





Manage Jenkins



Tools

Configure tools, their locations and automatic installers.



Scroll down until you find the Maven section

Maven

Maven installations



Edited



Maven installations

List of Maven installations on this system

Add Maven

Maven
Name

M3

Type



Install automatically ?

Install from Apache

Version

3.9.3

Add Installer

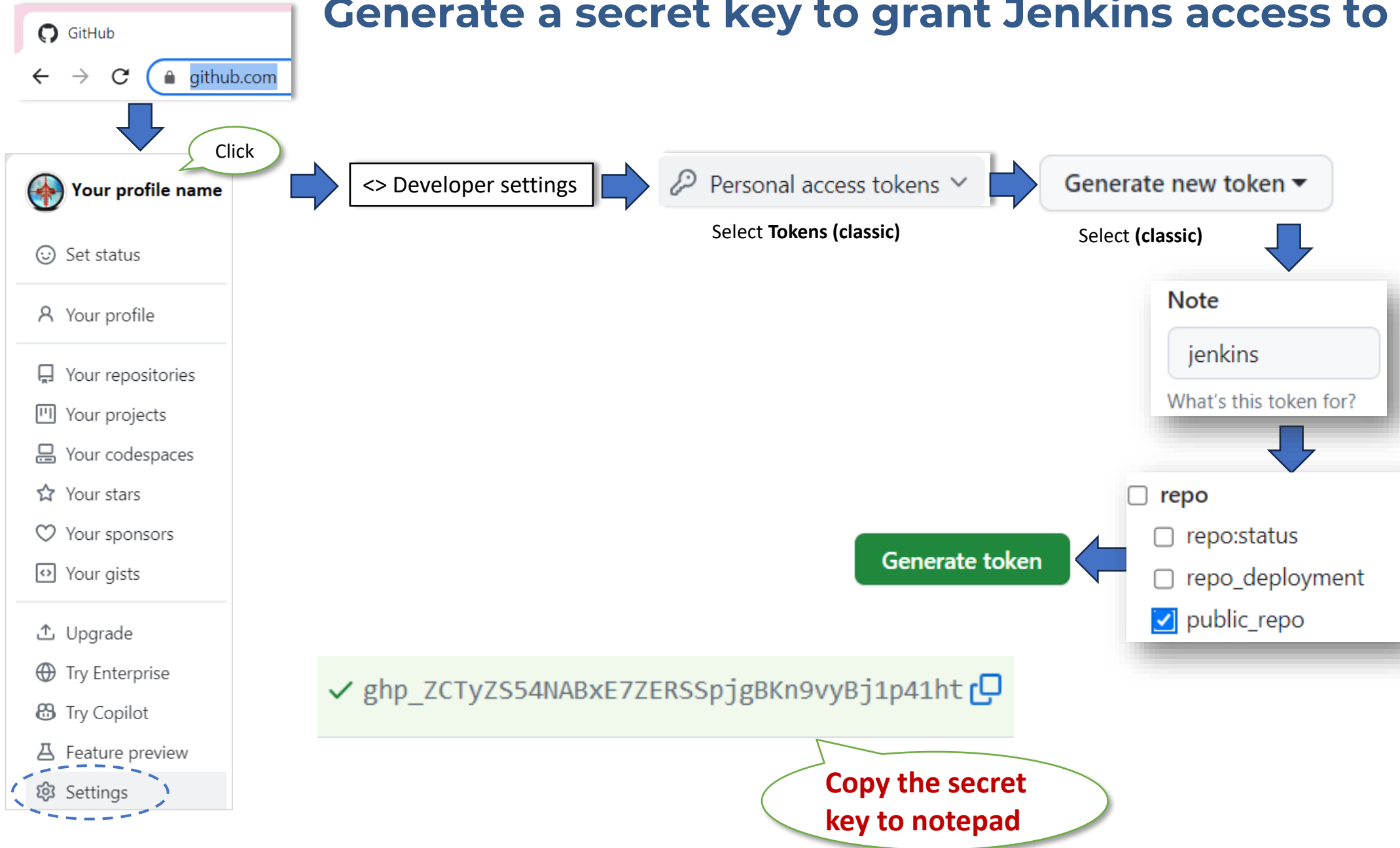
Add Maven

Click

Save

Apply

Generate a secret key to grant Jenkins access to GitHub



Create a repository for your code

1. Copy the **calculator-jenkins** folder from the **Jenkins** folder to **C:\calculator-Jenkins**
2. Open a terminal window, and navigate to C:\calculator-Jenkins
`cd c:\calculator-jenkins`
3. Initialise the folder as a new Git repo: `git init`
4. Add all of the files: `git add .`
5. Commit the files: `git commit -m "Initial commit"`
6. Create a new repository in GitHub and call it **calculator-Jenkins**
7. Type `git branch -M main` (in the terminal window)
`git remote add origin https://github.com/.../calculator-jenkins.git`
`git push -u origin main`



Copy the address



Create a new Jenkins job

+ New Item

Type

Calculator job

» Required field

Select

Freestyle project
This is the central feature of Jenkins

Maven project
Build a maven project. Jenkins

Click

OK

On the screen below

Source Code Management

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/mikebaradaran/calculator-jenkins.git

Continue on the next slide...



Credentials ?

- none -

Add ▲

Jenkins

Advanced ▼



Add Credentials

Domain

Global credentials (unrestricted) ▼

Kind

Secret text ▼

Scope ?

Global (Jenkins, nodes, items, all child items, etc) ▼

Secret

.....

ID ?

Description ?

Add Cancel



Branches to build ?

Branch Specifier (blank for 'any') ?

*/main

Save

Click the
arrow

Then
Click

Personal access token
from GitHub

Select



▶ Build Now



Build History

trend ▼



Filter builds...



#1

17 Jul 2023, 23:09



Atom feed for all



Atom feed for failures

Click to
see log



Console Output

Started by user [admin](#)

Running as SYSTEM

Building in workspace `/var/jenkins_home/workspace/Calculator` job

The recommended git tool is: NONE

No credentials specified

Cloning the remote Git repository

Cloning repository <https://github.com/mikebaradaran/calculator-jenkins.git>

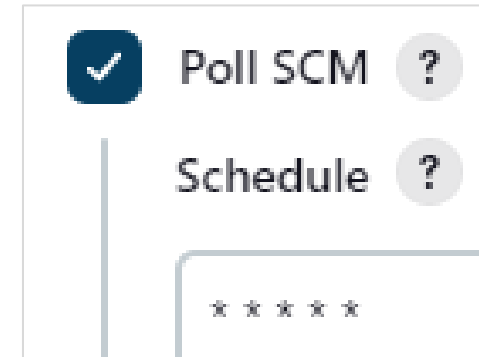
> `git init /var/jenkins_home/workspace/Calculator` job # timeout=10

Fetching upstream changes from <https://github.com/mikebaradaran/calculator-jenkins.git>



Configure Jenkins' Build Triggers

1. In the **Calculator job** window, select  **Configure**
2. Click on  **Build Triggers**
3. Check the **Poll SCM** (Source Control Management) box
Enter * * * * * into the **Schedule**
to build every minute every time a commit is made to this repo



The screenshot shows the Jenkins 'Build Triggers' configuration section. It includes a 'Poll SCM' checkbox which is checked, and a 'Schedule' field containing the text '* * * * *'. Both the checkbox and the schedule field have a help icon (a question mark in a circle) to their right.

Let's test this on the next page

Run the Trigger (Terminal window)

1. In the C:\ drive `/src/test/java/com/agile/calculator/CalculatorTest.java`
2. Change the value of **expected** on line 17 to **42** (**makes the test fail**)
3. Save the file
4. Add and commit this file to the repo

git add .

git commit -m "change test result"

git push

5. **Wait for a minute** (resist the urge to press the Build now button!)

This time the build fails, and the log indicates the reason for this failure

 #2	<u>17 Jul 2023, 23:32</u>
 #1	<u>17 Jul 2023, 23:09</u>